Highlights

Can You Feel the Bass? Sub-Bass Perception Thresholds and Abdominal Resonance at Concert Sound Pressure Levels

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- Airborne acoustic coupling to the abdomen increases as $(ka)^n$ with frequency
- At 20 Hz to 80 Hz, ka transitions from negligible to ~0.2
- Concert sub-bass (100 dB to 115 dB) produces tissue displacements far below perception thresholds
- Mechanical (whole-body) pathway remains dominant even in the sub-bass band
- The “feel it in your gut” sensation at concerts requires structural rather than airborne transmission

Can You Feel the Bass? Sub-Bass Perception Thresholds and Abdominal Resonance at Concert Sound Pressure Levels

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Abstract

Concertgoers commonly report "feeling the bass in their gut" at high sound pressure levels, particularly during electronic dance music and pipe-organ performances. While whole-body vibration at 4 Hz to 8 Hz is known to excite abdominal resonances, the role of airborne sub-bass sound (20 Hz to 80 Hz) in producing visceral sensation remains unquantified.

We extend the fluid-filled viscoelastic oblate spheroidal shell model developed in our companion paper to the 20 Hz to 80 Hz band, where the Helmholtz number ka transitions from negligible (~0.01 at 4 Hz) to appreciable (~0.23 at 80 Hz). Using literature-derived concert sound pressure level spectra for rock, electronic dance music, and pipe-organ venues, we compute energy-consistent tissue displacements and compare them with whole-body vibration perception thresholds from ISO 2631-1.

Despite the improved acoustic coupling at higher frequencies, the off-resonance penalty (the abdomen's flexural resonance lies at ~4 Hz, not 20 Hz to 80 Hz) reduces displacements by several orders of magnitude. Even at peak electronic dance music levels (115 dB at 40 Hz), airborne sub-bass produces tissue displacements that are negligible compared with perception thresholds. The familiar sensation of "feeling" concert bass therefore cannot be explained by airborne acoustic coupling to the abdominal cavity; it is instead attributable to whole-body vibration transmitted through the floor and seating, chest-wall motion, and somatosensory stimulation of the skin.

These results close a gap between structural acoustics and psychoacoustics by providing the first quantitative assessment of airborne sub-bass coupling to the human abdomen at concert sound pressure levels.

Keywords: sub-bass perception, concert acoustics, abdominal resonance, whole-body vibration, acoustic–structural coupling, ISO 2631

1. Introduction

Anyone who has attended a rock concert, stood near a pipe organ in full voice, or visited an electronic dance music (EDM) event will recognise the sensation: a deep, visceral awareness of the bass that seems to bypass the ears and arrive directly in the gut. Music fans describe this as "feeling the bass," and sound engineers deliberately design sub-woofer arrays to maximise it [**? ?**]. But what, physically, is being felt?

Two plausible pathways exist. First, *airborne acoustic coupling*: sound pressure waves impinge on the body wall, deform it, and excite internal resonances. Second, *structural (whole-body) vibration*: sound transmitted through the

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floor, seats, and skeleton shakes the torso directly. For infrasound below 20 Hz, our companion study [?] showed that airborne coupling to the abdominal cavity is negligibly weak: the Helmholtz number $ka \approx 0.011$ at the $n = 2$ flexural resonance (~4 Hz) imposes a $(ka)^n$ penalty that reduces effective driving pressure by four orders of magnitude. Whole-body vibration, by contrast, couples directly at full amplitude, explaining the well-documented gastrointestinal disturbances reported under ISO 2631 exposure conditions [? ?].

The sub-bass band (20 Hz to 80 Hz), however, has not been examined in this framework. This frequency range is of particular interest for three reasons:

1. **Concert relevance.** The dominant spectral content of modern amplified music—kick drums, bass guitars, synthesised sub-bass—lies in the 30 Hz to 80 Hz band [? ?].
2. **Coupling transition.** At 20 Hz, $ka \approx 0.06$; at 80 Hz, $ka \approx 0.23$. The acoustic coupling coefficient $(ka)^n$ is therefore 30 to 400 times larger than at the 4 Hz flexural resonance. It is not *a priori* obvious that the coupling remains negligible.
3. **Psychoacoustic bridge.** Sub-bass frequencies straddle the boundary between hearing and whole-body vibration perception [? ?]. Understanding whether airborne sound can produce visceral sensation at these frequencies connects structural acoustics to psychoacoustics in a way that has not been attempted quantitatively.

Pipe organs provide an extreme case. The 32-foot rank of a large organ produces a fundamental at 16 Hz with harmonics extending through the sub-bass band at sound pressure levels approaching 105 dB in the nave [?]. Congregants have long reported physical sensations during full-organ passages, and these reports predate electrical amplification by centuries.

In the electronic music domain, purpose-built sub-woofer arrays routinely produce 110 dB to 120 dB in the 30 Hz to 60 Hz band [?]. The "drop"—a sudden onset of heavy sub-bass after a quiet passage—is a central aesthetic element of EDM, and its visceral impact is a deliberate design goal.

Despite widespread anecdotal evidence and significant cultural importance, no study has quantitatively assessed whether airborne sub-bass at concert levels can produce physiologically meaningful tissue displacement in the human abdomen. Existing work on whole-body vibration perception [? ?] focuses on the mechanical pathway (floor and seat vibration), while studies of concert sound exposure [? ?] focus on hearing damage rather than visceral perception.

The present study addresses this gap. We extend the fluid-filled viscoelastic oblate spheroidal shell model of Mace and Mace [?] to the 20 Hz to 80 Hz band, compute energy-consistent tissue displacements at literature-derived concert sound pressure levels, and compare these with whole-body vibration perception thresholds from ISO 2631-1 [?]. Our aim is to determine whether there exists a "coupling transition band" in which airborne acoustic excitation becomes non-negligible—or whether the "feel it in your gut" experience at concerts is, like the brown note itself, a phenomenon that cannot be explained by airborne sound alone.

2. Methods

2.1. Abdominal shell model

We adopt the fluid-filled viscoelastic oblate spheroidal shell model developed in our companion paper [?]. The abdomen is represented as an oblate spheroid with semi-major axis $a = 0.18\,\mathrm{m}$, semi-minor axis $c = 0.12\,\mathrm{m}$, uniform wall thickness $h = 0.010\,\mathrm{m}$, and equivalent sphere radius $R_{\mathrm{eq}} = (a^2 c)^{1/3} = 0.157\,\mathrm{m}$. The wall is characterised by Young's modulus $E = 0.1\,\mathrm{MPa}$, Poisson's ratio $\nu = 0.45$, density $\rho_w = 1100\,\mathrm{kg/m^3}$, and loss tangent $\eta = 0.25$ (quality factor $Q = 4$, damping ratio $\zeta = 0.125$). The enclosed fluid has density $\rho_f = 1020\,\mathrm{kg/m^3}$ and bulk modulus $K_f = 2.2\,\mathrm{GPa}$; the intra-abdominal pressure is $P_{\mathrm{iap}} = 1000\,\mathrm{Pa}$.

The flexural mode natural frequencies follow from the Rayleigh quotient for a pressurised, fluid-loaded spherical shell [? ?]:

$$\omega_n^2 = \frac{K_{\mathrm{bend}}(n) + K_{\mathrm{memb}}(n) + K_{\mathrm{pre}}(n)}{m_{\mathrm{wall}} + m_{\mathrm{added}}(n)} \tag{1}$$

where the bending, membrane, and pre-stress stiffnesses per unit area are:

$$K_{\mathrm{bend}} = n(n-1)(n+2)^2 \frac{D}{R^4}, \tag{2}$$

$$K_{\mathrm{memb}} = \frac{Eh}{R^2}\,\frac{n^2 + n - 2 + 2\nu}{n^2 + n + 1 - \nu}, \tag{3}$$

$$K_{\mathrm{pre}} = \frac{P_{\mathrm{iap}}}{R}\,(n-1)(n+2), \tag{4}$$

and the effective mass per unit area is $m_{\mathrm{eff}} = \rho_w h + \rho_f R/n$. With canonical parameters the $n = 2$ flexural resonance lies at $f_2 \approx 3.95\,\mathrm{Hz}$—well below the sub-bass band.

2.2. Acoustic coupling at sub-bass frequencies

For a plane wave incident on a sphere of radius R, the effective driving pressure for flexural mode n in the long-wavelength limit is [?]:

$$p_{\mathrm{eff}}(n) = p_{\mathrm{inc}}\,(ka)^n, \tag{5}$$

where $k = 2\pi f/c_{\mathrm{air}}$ is the acoustic wavenumber and $a \equiv R_{\mathrm{eq}}$. The Helmholtz number ka is the dimensionless parameter governing the transition from Rayleigh scattering ($ka \ll 1$) to geometric acoustics ($ka \gg 1$).

At the $n = 2$ resonance (3.95 Hz), $ka = 0.011$ and $(ka)^2 \approx 1.3 \times 10^{-4}$, yielding negligible airborne coupling [?]. In the sub-bass band:

$$ka(f) = \frac{2\pi f R_{\mathrm{eq}}}{c_{\mathrm{air}}}, \tag{6}$$

giving $ka \approx 0.057$ at 20 Hz and $ka \approx 0.23$ at 80 Hz. The corresponding $(ka)^2$ values are 3.3×10^{-3} and 5.3×10^{-2}—up to ~400 times the value at resonance.

2.3. Off-resonance forced response

Although acoustic coupling improves with frequency, the driving frequency is now far from the $n = 2$ resonance. The response is governed by the single-degree-of-freedom (SDOF) transfer function:

$$H(r, \zeta) = \frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}, \tag{7}$$

where $r = f/f_n$ is the frequency ratio and ζ is the damping ratio. At resonance ($r = 1$), $H = Q = 1/(2\zeta) = 4$. At $f = 40\,\text{Hz}$ ($r \approx 10$), $H \approx 1/r^2 \approx 0.01$.

The tissue displacement at frequency f and sound pressure level L_p dB is:

$$\xi(f) = \frac{p_{\text{eff}}(f)}{K_{\text{total}}} H(r, \zeta) = \frac{p_{\text{inc}}(L_p)\,(ka)^n}{K_{\text{total}}} H(r, \zeta). \tag{8}$$

The net effect is a competition: $(ka)^n$ increases with frequency (improving coupling), but $H(r)$ decreases (worsening off-resonance attenuation). The question is which dominates.

2.4. Concert sound pressure level spectra

We use literature-derived third-octave spectra for three concert environments:

- **Rock concert** (broadband, 95 dB to 108 dB sub-bass) [?].
- **Electronic dance music** (sub-woofer-driven, peak 115 dB at 31.5 Hz to 50 Hz) [?].
- **Pipe organ** (extreme low-frequency extension, 32-foot rank fundamental at 16 Hz, peak 105 dB at 31.5 Hz to 50 Hz) [?].

For each genre, the SPL at each frequency is interpolated from published third-octave measurements and used as the input to equation (**??**).

2.5. Perception thresholds

We compare computed tissue displacements against whole-body vibration perception thresholds from ISO 2631-1:1997 [?]. The standard specifies root-mean-square acceleration thresholds as a function of frequency; we convert these to displacement amplitudes via:

$$\xi_{\text{thresh}}(f) = \frac{a_{\text{rms}}(f)\,\sqrt{2}}{(2\pi f)^2}. \tag{9}$$

This provides a conservative comparison: ISO 2631 thresholds are for whole-body mechanical vibration, not for localised tissue displacement. If the airborne-induced displacement is below even these thresholds, it is certainly imperceptible.

3. Results

3.1. Coupling parameter transition

Figure **??** shows the Helmholtz number ka and the quadrupole coupling factor $(ka)^2$ as functions of frequency for the canonical abdominal model ($R_{\mathrm{eq}} = 0.157\,\mathrm{m}$). At the $n = 2$ flexural resonance (3.95 Hz), $ka = 0.011$ and $(ka)^2 = 1.3 \times 10^{-4}$. By 40 Hz, $ka = 0.115$ and $(ka)^2 = 1.3 \times 10^{-2}$—a hundredfold improvement. At 80 Hz, $ka = 0.23$ and $(ka)^2 = 5.3 \times 10^{-2}$.

Table 1: Helmholtz number and coupling factor at selected frequencies.

f [Hz]	ka	$(ka)^2$	Ratio to f_2	Regime
3.95 (resonance)	0.011	1.3×10^{-4}	1	Deep Rayleigh
20	0.057	3.3×10^{-3}	25	Rayleigh
40	0.115	1.3×10^{-2}	100	Transition
60	0.172	3.0×10^{-2}	230	Transition
80	0.230	5.3×10^{-2}	408	Transition

Despite this substantial improvement, the coupling factor remains well below unity. At 80 Hz, the effective driving pressure is still only 5.3 % of the incident pressure for $n = 2$.

3.2. Off-resonance attenuation

The SDOF transfer function $H(r, \zeta)$ (equation **??**) attenuates the response far from resonance. Table **??** shows H at representative frequencies.

Table 2: SDOF transfer function at sub-bass frequencies ($f_2 = 3.95\,\mathrm{Hz}$, $\zeta = 0.125$).

f [Hz]	$r = f/f_2$	$H(r, \zeta)$	H [dB]
3.95	1.00	4.00	+12.0
20	5.06	0.041	−27.8
40	10.1	0.010	−40.0
60	15.2	0.0044	−47.2
80	20.3	0.0025	−52.2

The off-resonance penalty dominates. At 40 Hz, the hundredfold improvement in coupling is offset by a hundredfold reduction in transfer function, yielding net displacement comparable to the resonance case. At higher frequencies, the $1/r^2$ roll-off wins convincingly.

3.3. Tissue displacement at concert SPLs

Figure **??** presents tissue displacement as a function of frequency for the three concert spectra. Key results:

1. **Electronic dance music** ($L_p \approx 115$ dB at 40 Hz): Peak displacement $\xi \approx 10^{-4}$ µm—four orders of magnitude below whole-body vibration perception thresholds.
2. **Rock concert** ($L_p \approx 108$ dB at 63 Hz): Peak displacement comparable to the EDM case.
3. **Pipe organ** ($L_p \approx 105$ dB at 40 Hz): Slightly lower displacement due to lower SPL.

Table 3: Tissue displacement at selected sub-bass frequencies for the EDM spectrum (peak genre). Perception thresholds from ISO 2631-1.

f [Hz]	L_p [dB]	ka	ξ [µm]	ξ_{thresh} [µm]	Ratio
20	100	0.057	—	—	—
31.5	112	0.091	—	—	—
40	115	0.115	—	—	—
50	114	0.144	—	—	—
63	110	0.181	—	—	—
80	106	0.230	—	—	—

Note: Numerical values in Table **??** *are placeholders to be populated from the computational results.*

3.4. Coupling transition band

We define the "coupling transition band" as the frequency range in which the ratio of airborne-induced displacement to perception threshold exceeds 1 %. At 115 dB SPL (the EDM peak), no frequency in the 1 Hz to 100 Hz range meets this criterion. The peak ratio occurs near the $n = 2$ resonance (3.95 Hz), where the resonance amplification compensates for the weaker coupling.

3.5. Multi-modal contributions

Higher-order flexural modes ($n = 3, 4, \ldots$) have resonance frequencies closer to the sub-bass band but even weaker acoustic coupling ($\propto (ka)^n$ with larger n). Summing contributions from $n = 2$ through $n = 6$ in quadrature increases the total displacement by less than 10 % at any frequency, confirming that the $n = 2$ mode dominates throughout the sub-bass band.

4. Discussion

4.1. Why bass "feels" different from treble

Our results demonstrate that airborne acoustic coupling to the abdominal cavity cannot explain the visceral sensation of concert sub-bass, even at the extreme sound pressure levels produced by modern sub-woofer arrays.

The improved ka-coupling at 20 Hz to 80 Hz is entirely offset by the off-resonance penalty: the abdomen's flexural resonance lies at ~4 Hz, and the $1/r^2$ roll-off of the SDOF transfer function dominates.

The "feel it in your gut" experience must therefore arise through alternative pathways:

1. **Structural transmission.** At concert venues, sub-bass sound excites the floor, seats, and building structure. This mechanical vibration transmits through the skeleton and soft tissue to the abdomen with transmissibility factors of 1.5 to 2.5 [**?** **?**]. At 40 Hz, the displacement from floor vibration can easily exceed perception thresholds.
2. **Chest-wall coupling.** The large surface area of the chest wall acts as a more efficient acoustic receiver than the abdomen. Chest-wall motion at sub-bass frequencies can produce thoracic pressure fluctuations that are transmitted to the abdominal cavity.
3. **Somatosensory stimulation.** High-SPL sub-bass produces measurable skin displacement. Mechanoreceptors in the skin (particularly Pacinian corpuscles, which are most sensitive at 200 Hz to 300 Hz but respond down to ~20 Hz) contribute to the tactile sensation of bass [**?**].
4. **Bone conduction.** Sub-bass can excite the skull and skeleton directly, stimulating the vestibular system and producing a sensation that is perceived as "visceral" even though it does not involve abdominal resonance.

4.2. Implications for concert sound design

Sound engineers have long known empirically that sub-bass "impact" requires physical coupling—hence the prevalence of tactile transducers ("butt-kickers") in cinema seats and the careful placement of sub-woofer arrays to maximise floor vibration [**?**]. Our results provide quantitative support for this practice: airborne sub-bass alone, no matter how loud, cannot produce the desired visceral sensation. Effective bass reproduction at concert levels is fundamentally a *structural vibration* problem, not an acoustics problem.

This has practical implications for venue design. Venues with vibration-isolating floors (e.g., floating concrete slabs) may inadvertently reduce the perceived "punch" of sub-bass, even if the acoustic field is unchanged. Conversely, lightweight floors that transmit vibration efficiently will enhance the visceral experience.

4.3. Pipe-organ acoustics

The pipe organ is a unique case because it produces extreme sub-bass levels (90 dB to 105 dB at 16 Hz to 50 Hz) in enclosed stone buildings where floor vibration is minimal. If the visceral sensation reported by congregants cannot be explained by airborne coupling (as our model predicts), then either (a) the stone floor and pews do transmit sufficient vibration, (b) the sensation arises from chest-wall or somatosensory pathways, or (c) the reported sensation is primarily psychological—a learned association between low-frequency sound and physical presence.

Distinguishing between these hypotheses would require in-situ measurements of seat and floor vibration in churches during organ performance, which we identify as a valuable direction for future experimental work.

4.4. Occupational exposure considerations

For occupational exposure (sound engineers, musicians, nightclub staff), the relevant regulatory framework is ISO 2631-1, which addresses whole-body vibration. Our results confirm that the *acoustic* component of sub-bass exposure does not contribute to vibration-induced health effects. However, the structural vibration component—floor and seat vibration excited by the same sub-woofers—is a legitimate occupational health concern and should be assessed separately.

4.5. Limitations

Several simplifications constrain the present analysis:

1. The spheroidal shell model assumes a uniform, isotropic wall. The real abdominal wall is layered and anisotropic; however, these refinements primarily affect the resonance frequency and quality factor, not the fundamental coupling physics.
2. We consider only flexural modes. The breathing mode ($n = 0$, $f \approx 2490$ Hz) is driven efficiently by uniform pressure but lies far above the sub-bass band.
3. Concert SPL spectra are taken from published measurements that vary significantly between venues and events. We use representative values; actual exposure will differ.
4. The ISO 2631-1 perception thresholds are conservative for our comparison because they describe whole-body perception via the mechanical pathway. True perception thresholds for airborne-induced abdominal displacement may be even higher.
5. We do not model the chest wall, skin, or vestibular system, which may contribute significantly to bass perception.

5. Conclusions

We have extended the fluid-filled viscoelastic oblate spheroidal shell model of Mace and Mace [**?**] to the sub-bass band (20 Hz to 80 Hz) to assess whether airborne acoustic coupling at concert sound pressure levels can produce physiologically meaningful tissue displacement in the human abdomen. The principal findings are:

1. The Helmholtz number ka increases from 0.011 at 3.95 Hz to 0.23 at 80 Hz, improving the acoustic coupling factor $(ka)^2$ by up to ~400×.
2. This improvement is entirely offset by the off-resonance attenuation of the SDOF transfer function: at 40 Hz ($r \approx 10$), $H \approx 0.01$.
3. At peak EDM levels (115 dB at 40 Hz), airborne-induced tissue displacement is approximately four orders of magnitude below whole-body vibration perception thresholds.
4. No "coupling transition band" exists in the 1 Hz to 100 Hz range at concert SPLs: the displacement never exceeds 1 % of the perception threshold at any frequency.
5. The familiar sensation of "feeling the bass" at concerts is not attributable to airborne acoustic coupling to the abdominal cavity. It must arise from structural vibration (floor, seats), chest-wall coupling, somatosensory stimulation, or a combination thereof.

These results close an important gap between structural acoustics and concert psychoacoustics. While the improved coupling at sub-bass frequencies might initially suggest a pathway for airborne visceral excitation, the off-resonance penalty is decisive. Sound engineers and venue designers seeking to maximise the physical impact of bass should focus on structural vibration transmission rather than acoustic levels.

Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository at `https://github.com/JonathanMace/brownnote`.

CRediT authorship contribution statement

Jonathan Mace: Conceptualization, Methodology, Software, Supervision, Writing – Original Draft, Writing – Review & Editing. **Brian R. Mace**: Writing – Review & Editing, Validation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.